

Aditya Vashisth

Engineering Lead

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Technical Skills

Core AI/ML & Agentic AI

- Generative AI: OpenAI GPT-4o, Gemini, Fine-tuning, Prompt Engineering
- Vertex AI, GCP, Agent Engine
- Multi-Agent Frameworks: Microsoft AutoGen, LangGraph.
- LLM Frameworks: LangChain, LlamaIndex, Semantic Kernel, LangSmith
- RAG Systems: VectorDB(Faiss, Milvus, Pinecone), Retrieval-Augmented Generation
- Agent Design: Conversational AI, Human-in-the-Loop, Autonomous Workflows, Tool Integration

Programming & Development

- Languages: Python (Primary), JavaScript, Java
- Web Technologies: React.js, Node.js, REST APIs, FastAPI, Streamlit
- Cloud Platforms: GCP, Azure AI Services, AWS and OpenAI API.

Data & Infrastructure

- Databases: Vector DB(Faiss, Pinecone and Milvus), MySQL and Firestore.
- MLOps: MLflow, Dataiku.
- Tools: Git, Agile SDLC, Azure AI Search, GCP Vector Search

Certifications

- Google Cloud Certified-Professional Cloud Architect
- Google Cloud Certified-Professional Machine Learning Engineer
- Google Cloud Certified-Professional Data Engineer
- Google Cloud Certified-Associate Cloud Engineer
- Google Cloud Certified-Cloud Digital Leader

Profile

Innovative AI Engineering Lead with 8+ years of experience specializing in Generative AI, Large Language Models, and Multi-Agent Systems. Expert in building production-ready AI solutions using OpenAI APIs, LangChain, LangGraph, and Microsoft AutoGen framework. Proven track record in designing autonomous agent workflows, implementing RAG systems, and orchestrating complex multi-agent conversational AI systems for enterprise applications.

Education

Guru Gobind Singh Indraprastha University, Delhi | 2018 - 21

Master of Technology (Computer Science)

Guru Gobind Singh Indraprastha University, Delhi | 2013 - 17

Bachelor of Technology (Computer Science)

Work Experience

Engineering Lead | Persistent Systems | Apr 2022 - Present

Generative AI & Multi-Agent Systems Leadership:

- Lead development of enterprise-grade Generative AI applications using OpenAI GPT-3.5-Turbo, LangChain.
- Architect and implement multi-agent conversational systems with specialized agent roles (AssistantAgent, UserProxyAgent, Custom Agents) for complex workflow automation
- Design autonomous agent ecosystems using LangGraph for dynamic workflow orchestration and decision-making
- Implement advanced RAG (Retrieval-Augmented Generation) systems with vector search capabilities using FAISS and Azure AI Search Service

Technical Architecture & Implementation:

- Build intelligent chatbots with context-aware conversation management and memory persistence
- Create secure middleware layers for AI API integration, authentication, and rate limiting
- Architect data processing pipelines for content extraction, transformation, and embedding generation
- Implement vector search and semantic similarity matching for enhanced AI response accuracy.

Software Engineer | TechTree IT Systems | Sep 2019 - Mar 2022

- Engineered features and enhancements for enterprise applications with a focus on performance optimization and cross-platform compatibility.

Software Engineer | Sagar Info. Pvt Ltd | Aug 2017 - Sep 2019

- Built and maintained robust software solutions with emphasis on application performance optimization and user experience improvements.

Key Projects

Heart Flow - Contract Intelligence Agent | Agentic AI System

Developed a sophisticated contract analysis and metadata extraction agent using Google Agent Development Kit (ADK) for enterprise legal document processing.

- Document Discovery & Pagination: Implemented intelligent document search with word-matching algorithms and pagination support, enabling efficient retrieval from large document repositories
- LLM-Based Metadata Extraction: Built an intelligent metadata extraction pipeline that processes PDF contracts and automatically identifies key entities: client names, service providers, dates, contract terms, and signatory information
- Smart Document Selection: Implemented multi-criteria document ranking logic that prefers final/executed versions over drafts, selects the highest version numbers, and prioritises most recent dates for accurate analysis
- GCS Integration: Seamlessly integrated Google Cloud Storage for document access and retrieval with robust error handling and format validation
- Scalable Architecture: Designed a pagination-based approach for handling large document collections (100K+ files) without performance degradation
- RAG-Ready Output: Generates structured JSON output with complete contract metadata for downstream RAG systems and compliance workflows

HealthEdge Guiding Care Bot | Multi-Agent System

Developed an advanced multi-agent AI assistant using Python, OpenAI, and LangChain for automated ticket management and customer support.

- Implemented conversational agent workflows with specialized roles for ticket classification, response generation, and escalation handling
- Developed responsive React.js UI components with real-time agent communication display
- Engineered Node.js middleware with advanced API security, authentication, and rate limiting
- Engineered data integration system incorporating Confluence pages, HTML dumps, and video content
- Enhanced search capabilities using Azure AI Search Service and advanced prompt systems

Regeneron Pharmaceuticals Target PPM Bot

Built a conversational AI bot using Python, LangChain, and OpenAI models

- Implemented the Milvus vector database for efficient embedding, storage, and semantic search
- Created end-to-end RAG pipeline from data ingestion to contextual response generation
- Deployed solution as ML-Flow model and converted to REST API using Dataiku Platform
- Implemented context-aware conversation management with memory persistence

Essay Writer with LangGraph Multi-Agent System

GitHub: github.com/av525/essay-writer-langgraph

Developed a multi-agent AI system using LangGraph for automated essay writing with specialized agent collaboration

- Implemented 5-agent workflow: Planner, Researcher, Writer, Critic, and Research Critic agents for iterative content improvement & Integrated Tavily API for real-time web research and Azure OpenAI for content generation.
- Built Streamlit web interface with real-time execution tracking and configurable revision cycles
- Designed state management system using LangGraph for persistent workflow orchestration

AI Customer Support with AutoGen Conditional Nested Chat

GitHub: github.com/av525/ai-customer-support

Architected sophisticated customer support application using AutoGen Conditional Nested Chat Architecture

- Implemented intelligent agent routing with 4 specialized agents: Support Coordinator, Response Agent, Troubleshooting Agent, and Escalation Agent
- Built conditional workflow system where agents are activated based on inquiry type and complexity
- Developed real-time processing visualization with live agent flow tracking and performance monitoring